

Sercan Özen

Solar Cell Engineer | Space Photovoltaics, Characterization & Reliability

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Final-year PhD researcher in Physics specializing in space photovoltaics, with experience in the fabrication of perovskite solar cells and the characterization and reliability assessment of perovskite, organic, and silicon single- and multi-junction devices. Directly contributed to two orbital solar-cell payloads, spanning device fabrication, characterization, encapsulation, electrical contacting, and PCB integration. Experienced in advanced optoelectronic diagnostics and extreme-environment testing, including low-intensity, low-temperature operation, thermal cycling, and in situ and in operando proton irradiation. Develops automated measurement and data-analysis tools to identify device-level performance losses and degradation mechanisms.

Core R&D Expertise

- Space Photovoltaics and Extreme-Environment Reliability
- Proton Irradiation and Radiation-Induced Degradation
- Space-Compatible Encapsulation and Integration
- Single-, Tandem-, and Triple-Junction Solar Cells
- Optoelectronic Characterization and Device Physics
- Experimental Setup Development and Measurement Automation

Further details on research projects, experimental systems, and spaceflight work: srcnzn.github.io

Professional Experience

Doctoral Researcher, University of Potsdam 2023–Present

- Develop and characterize perovskite-based single-, tandem-, and triple-junction solar cells for space applications.
- Design and execute in situ and in operando 68 MeV proton-irradiation experiments in the dark and under AM0 illumination.
- Develop and assess space-compatible device packaging approaches, including SiO_x encapsulation, adhesive-bonded cover-glass integration, electrical contacting, and PCB integration, for environmental testing and space applications.
- Investigate degradation, recovery, voltage losses, and subcell behavior using J–V characterization, MPP tracking, EQE, PL, PLQY, and EL measurements.
- Design automated measurement and analysis workflows using Python, LabVIEW, and MATLAB.

Spaceflight Missions

- **CyBEEsat** — launched aboard the Isar Aerospace Spectrum launch vehicle September 2026
Payload: ROSI-1 perovskite solar cells
Executed the complete device-to-payload workflow, including solar-cell fabrication, characterization, encapsulation, electrical contacting, and PCB integration.
- **OOV-Cube** — launched aboard Ariane 6 July 2024
Payload: Perovskite/CIGS and perovskite/silicon tandem solar cells
Performed pre-flight characterization, encapsulation, electrical contacting, PCB integration, and electrical testing; initial in-orbit data confirmed power generation under non-ideal solar alignment.

Research Assistant, Izmir Institute of Technology 2018–2023

- Built, calibrated, and maintained optical laboratory setups while supporting laboratory teaching and materials/device research.

Graduate Researcher (MSc), CENT Research Group 2017–2020

- Synthesized and doped perovskite materials and investigated their optical and electronic properties.

Engineering Intern, DVLX R&D 2015–2016

- Designed and assembled modular electric-bicycle conversion prototypes, including battery packs, mechanical holders, and motor-controller programming.

Education

PhD in Physics, University of Potsdam 2023–2027 (expected)

Dissertation (provisional): *Understanding Operation and Degradation Mechanisms in Single- and Multi-Junction Solar Cells for Space Applications*

MSc in Photonics Science and Engineering, Izmir Institute of Technology 2017–2020

Thesis: *Experimental and Theoretical Investigation of Functionalized Perovskites*

BSc in Physics, Izmir Institute of Technology 2017

Languages

Turkish — Native | English — Full professional proficiency | German — A1; currently studying at A2 level

Technical Expertise

PV Device Fabrication & Spaceflight Integration

Single- and multi-junction perovskite solar-cell fabrication using glovebox-based solution processing and spin coating; thermal evaporation of metal contacts; layer-stack engineering; device packaging, including SiO_x encapsulation and adhesive-bonded cover-glass integration; electrical contacting; PCB integration and electrical testing.

Space PV Reliability & Extreme-Environment Testing

Solar-cell operation and stability testing under AM0 illumination; low-intensity, low-temperature (LILT) testing; thermal cycling; fluctuating irradiance; in situ and in operando proton irradiation; degradation and recovery analysis; pre-flight validation and in-orbit performance assessment.

Electrical & Optoelectronic Characterization

Characterization of perovskite, silicon, and organic single-junction solar cells, as well as perovskite/organic and perovskite/silicon tandem devices. Temperature- and intensity-dependent J–V characterization; MPP tracking; EQE/IQE; subcell-resolved multi-junction analysis; resistance-dependent photovoltage (RPV); steady-state, intensity-dependent, and temperature-dependent PL; spatially resolved PL and EL imaging and mapping; PLQY and ELQY.

Experimental Systems & Calibration

Design, assembly, calibration, and automation of custom J–V, EQE, PLQY, and ELQY setups, as well as in situ and in operando proton-irradiation measurement systems.

Data Analysis & Automation

Custom software for J–V, stability, EQE, PLQY, and signal/background data processing and analysis; instrument control; automated measurement workflows; photovoltaic data visualization and interpretation. Python (PyQt/PySide), LabVIEW, MATLAB, Origin, and Git.

Materials Characterization

XRD, GI-XRD, GIWAXS, and SEM for structural and morphological characterization of thin films and photovoltaic devices.

Selected Publications & Manuscripts

All-Perovskite Triple-Junction Solar Cells for Space: In Operando Response to Proton Irradiation

S. Ozen, K. J. Prince, Y. Yun, B. Li, et al. *Materials Horizons*, manuscript under revision, 2026.

Performance Constraints of All-Perovskite Tandem Solar Cells in Low-Intensity, Low-Temperature Environments

S. Ozen, E. Beier, F. Peña-Camargo, J. Thiesbrummel, G. Boccarella, et al. *Advanced Materials*, 38(15), e17703, 2026.

Beyond Earth: Resilience of Quasi-2D Perovskite Solar Cells in Space

C. Putz, L. E. Lehner, S. Demchyshyn, B. Hailegnaw, P. Jahelka, M. Breitwieser, S. Ozen, et al. *Advanced Materials*, 38(31), e20433, 2026.

Understanding and Mitigating Atomic Oxygen-Induced Degradation of Perovskite Solar Cells for Near-Earth Space Applications

B. A. Seid, S. Sarisozen, F. Peña-Camargo, S. Ozen, E. Gutierrez-Partida, et al. *Small*, 20(30), 2311097, 2024.

Methylammonium-free Co-evaporated Perovskite Absorbers with High Radiation and UV Tolerance: An Option for In-space Manufacturing of Space-PV?

F. Lang, Y.-H. Chiang, K. Frohna, S. Ozen, H. C. Neitzert, et al. *RSC Advances*, 13(31), 21138–21145, 2023.

Moon Photovoltaics Utilizing Lunar Regolith and Halide Perovskites

J. M. Cuervo-Ortiz, J. C. G. Palomares, S. Ozen, M. Härtel, S. Sarisozen, et al. *Device*, 3(7), 100747, 2025.

Full publication list: Google Scholar profile.

Selected Conferences

3rd Radiation Hardness Testing Workshop, Berlin, Germany

2026

In Situ and In Operando Performance of 3J-PSCs Under High-Energy Proton Irradiation; poster presentation.

PVSPACE-25, Rome, Italy

2025

Understanding Performance and Stability of Perovskite-Based Multi-Junction Solar Cells for Space Applications; invited speaker.

4th International tandemPV Workshop, Amsterdam, the Netherlands

2024

All-Perovskite Tandem Solar Cells for Low-Intensity, Low-Temperature Environments; poster presentation.

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